The Double-Slit Experiment: Why Interference Is the Expected Default and Non-Interference Requires Explanation

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Abstract

We propose that the double-slit interference pattern is not a mysterious quantum phenomenon requiring special postulates. It is instead the straightforward, almost inevitable outcome when a localized entropy-density gradient packet propagates through a sparse region of the pre-geometric lattice, where residual primordial bonds remain active and couple both paths sub-locally. Non-interference occurs only when a measurement apparatus partitions those primordial bonds. This perspective follows directly from a single scalar ontology and unifies the double-slit with entanglement, retrocausality, and black-hole information preservation without introducing new equations or entities.

1 Introduction

The double-slit experiment is a canonical demonstration that quantum systems behave in ways that appear to defy classical intuition. Standard quantum mechanics accounts for the interference pattern with a wavefunction that passes through both slits simultaneously. Pilot-wave theory requires a real guiding wave in configuration space. Both approaches treat interference as the phenomenon that needs special explanation.

Here we show the opposite: interference is the default, expected behavior. Any loss of interference requires an additional physical process that suppresses sub-local correlations. This inversion arises naturally from the pre-geometric lattice ontology presented in EUT II [2] and the black-hole interior construction of EUT I V9 [1].

2 The Pre-Geometric Lattice

The underlying ontology is a bare index set $\{i\}$ of real scalar entropy densities S_i equipped with all-to-all entropic affinities

$$w_{ij} = \left[1 + \frac{(S_i - S_j)^2}{T_i T_j}\right]^{-\beta}, \quad \beta \simeq 3.$$

Spatial geometry emerges only after free-energy minimization at the rigidity-percolation transition ($\beta \approx 3$), pruning weaker bonds and inducing a 3D chart. The surviving high-stiffness bonds are the *residual primordial bonds*. These never participate in dimensional emergence and connect sites sub-locally, beneath the emergent metric.

A "particle" (electron, photon, etc.) is a localized gradient packet of entropy density propagating through this lattice. In the single-particle regime the emergent spatial graph is extremely sparse, so the primordial layer remains highly relevant.

3 Single-Slit Propagation

With only one slit open, the gradient packet follows the available spatial paths. The primordial bonds are present but do not produce visible interference because there is only one dominant route. The packet arrives at the screen with a single-slit diffraction envelope, exactly as observed.

4 Double-Slit: Interference as the Default

When both slits are open and no which-path measurement is performed, two spatial paths become available. Because the lattice is sparse, a large set of residual primordial bonds directly connect sites behind slit 1 to sites behind slit 2 (and onward to the screen). These bonds are sub-local.

The global Dirichlet energy is now minimized across *both* paths simultaneously:

$$E_{\text{spatial}} = \frac{1}{2} \sum_{i,j} w_{ij} (S_i - S_j)^2.$$

The entropy-density packet therefore evolves under the influence of both slits at once. On the detection screen this produces the characteristic interference pattern. No wavefunction in configuration space is required; the pattern is the direct result of ordinary lattice energy minimization when primordial bonds remain active.

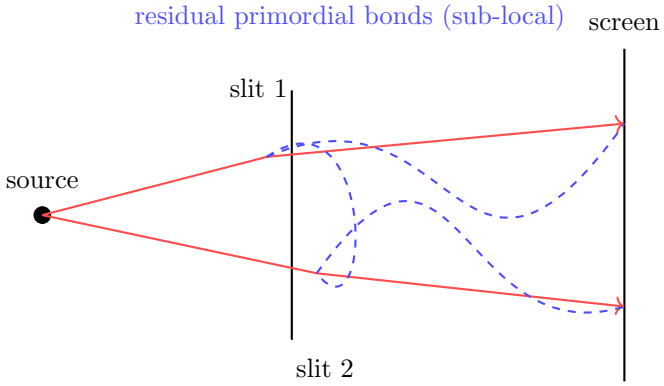

Figure 1: Schematic of the double-slit setup in the pre-geometric lattice. Red lines show emergent spatial paths through the two side-by-side slits on a single barrier. Blue dashed lines represent residual primordial bonds that couple the two paths sub-locally, producing interference as the default outcome of global Dirichlet energy minimization.

5 Which-Path Measurement: Why Non-Interference Requires Explanation

A which-path detector interacts locally with the packet near one slit. This raises the local temperature, which softens the bonds and lowers the effective rigidity, thereby triggering a partition of

the primordial bonds associated with the unchosen path. The sub-local correlations between the two paths are suppressed. The packet now propagates effectively along only one spatial path, and the interference pattern disappears.

Thus, the loss of interference is not the natural default. It requires an extra physical process: local partitioning of the primordial layer.

6 Unified Picture

The same sub-local primordial bond mechanism explains entanglement (EUT II §10), TSVF retrocausal back-reaction, and black-hole information preservation (EUT I V9, Appendix E). The double-slit is simply the simplest macroscopic demonstration of this deeper layer.

7 Conclusion

In this framework, double-slit interference is the expected, almost mundane outcome whenever a gradient packet propagates through a sparse lattice region with active primordial bonds. Non-interference occurs only when measurement deliberately partitions those bonds. The "weirdness" disappears; what remains is ordinary lattice dynamics.

The double-slit is not mysterious. It is what any entropy-density gradient packet would naturally do.

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References

[1] M. Steiniger, The Entropic Universe: An Effective Field Theory for Emergent Geometry and Localized Gradient Effects, Zenodo, doi:10.5281/zenodo.17528477 (2025–2026).

[2] M. Steiniger, The Entropic Universe II: Space, Time, Branching, and the Low-Entropy Past from a Single Scalar Line, Zenodo, doi:10.5281/zenodo.17651888 (2026).